DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech **Branch: Electronics & Telecommunication Engineering and Allied**

Subject Code & Name: BTETC401 Network Theory **Semester: IV**

Max Marks: 60 **Date: 19/12/2024** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 is compulsory.*
3. *Attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

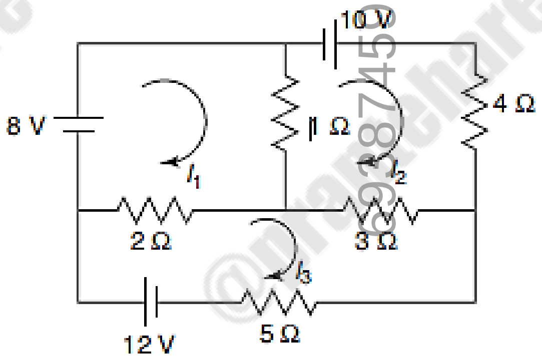
		(Level/CO)	Marks
Q.1	**Objective type questions. (Compulsory Question)**		**12**
1	The number of independent loops for a network with n nodes and b branches is. a. n − 1 b. b − n c. b − n + 1 d. Independent of the number of nodes	Understand	1
2	All high-frequency components are rejected by which of the following filter? a. Low Pass Filter b. High Pass Filter c. Band Pass Filter d. Band Stop Filter	Remember	1
3	A DC voltage source is connected across a series R-L-C circuit. Under steady-state conditions, the applied DC voltage drops entirely across the a. R only Understand c. C only d. R and L combination	Understand	1
4	The admittance and impedance of the following kind of network have the same properties a.LC b. RL c. RC d. RLC	Understand	1
5	The minimum amount of hardware required to make a low pass filter is a. a resistance, a capacitance and an op-amp. b. a resistance, an inductance and an op-amp. c. a resistance and a capacitance. d. a resistance, a capacitance and an inductance.	Remember	1
6	A two-port network is simply a network inside a black box, and the network has only a. two terminals b. two pairs of accessible terminals c. two pairs of ports d. All of the Above	Understand	1
7	When a series RL circuit is connected to a voltage V at t = 0, the current passing through the inductor L at t = 0+ is a. V/R b. infinite c. zero d. V/L	Understand	1

8	Superposition theorem is valid only for a. linear circuits b. non-linear circuits c. both linear and non-linear d. neither of the two	Remember	1
9	The Laplace transform of a unit step function is a. $1/S$ b. 1 c. $1/S^2$ d. $1/(S+a)$	Apply	1
10	The final value theorem is used to find the a. steady state value of the system output b. initial value of the system output c. transient behaviour of the system output d. none of these	Apply	1
11	If $Z_{11} = 2\ \Omega$; $Z_{12} = 1\ \Omega$; $Z_{21} = 1\ \Omega$ and $Z_{22} = 3\ \Omega$, what is the determinant of admittance matrix. a. 5 b. 1/5 c. 1 d. 0	Analyze	1
12	The nodal method of circuit analysis is based on a. KVL and Ohm's law b. KCL and Ohm's law c. KCL and KVL d. KCL, KVL and Ohm's law	Understand	1

Q.2 Solve the following. **12**

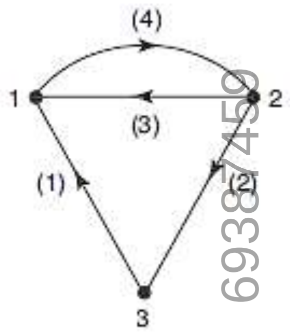
A) State and explain Kirchhoff's laws. **CO1** **6**

Also Determine the current through the 5 Ω resistor of the network shown using Mesh Analysis



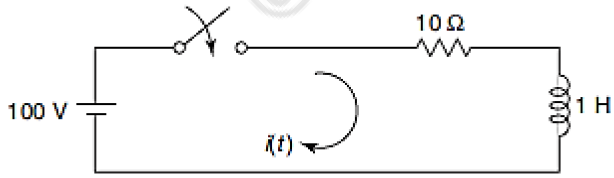
B) Define the terms: Sub-graph and Planar Graph. **CO1** **6**

Also, for the oriented graph shown write the (a) incidence matrix, (b) tie set matrix

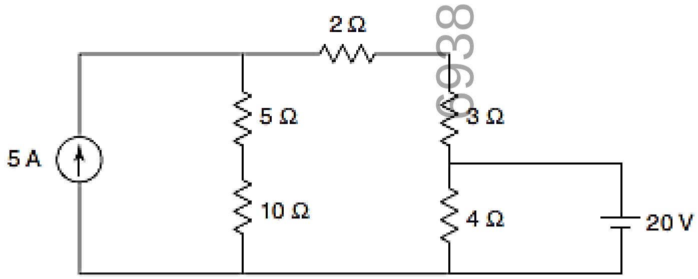


Q.3 Solve the following. **12**

A) In the given network, the switch is closed at t = 0. With zero current in the inductor, find i, di/dt, d^2i/dt^2 at $t = 0^+$. **CO1, CO3** **6**

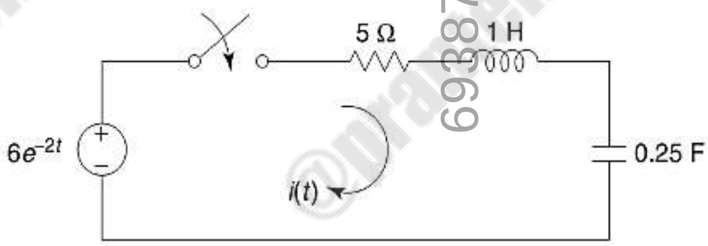


B) State Superposition Theorem and Find the current through the 3 Ω resistor in the fig. shown below **CO1** **6**



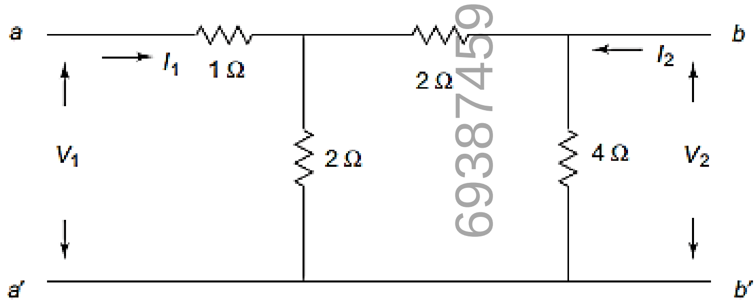
Q.4 Solve Any Two of the following. **12**

A) For the network shown above, the switch is closed at t =0. Determine the current i(t) assuming zero initial conditions in the network elements **CO1, CO3** **6**



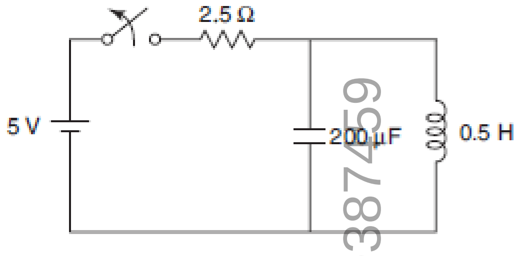
B) Find Laplace Transform of various standard time domain functions. **CO1** **6**

C) Define Driving-point Impedance and Driving-point Admittance Function. Also Find the h parameters of the network shown **CO1, CO4** **6**



Q.5 Solve Any Two of the following. **12**

A) In the network of Fig. Shown, the switch is closed and steady-state is attained. At t > 0, switch is opened. Determine the current through the inductor **CO1, CO3** **6**



B) Obtain the transient current and voltage responses of a RL circuit when subjected to a unit step input **CO1, CO3** **6**

C) Test Whether Polynomial given below are Hurwitz **CO1, CO3** **6**

i. $P(S) = S^4 + 7S^3 + 6S^2 + 21S + 8$

ii. $P(S) = S^4 + S^3 + 2S^2 + 3S + 2$

Q.6 Solve Any Two of the following. **12**

A) Explain about classification of filters. Draw the characteristics of Low-pass and High-pass filters. **CO1, CO3** **6**

B) Realize the following function in Cauer-I and Cauer-II forms **CO1, CO3** **6**

$$Z(S) = \frac{(S+1)(S+3)}{S(S+2)}$$

C) Derive expression of Z-parameters in Terms of y-parameters and Vice-versa **CO1, CO4** **6**

***** End *****